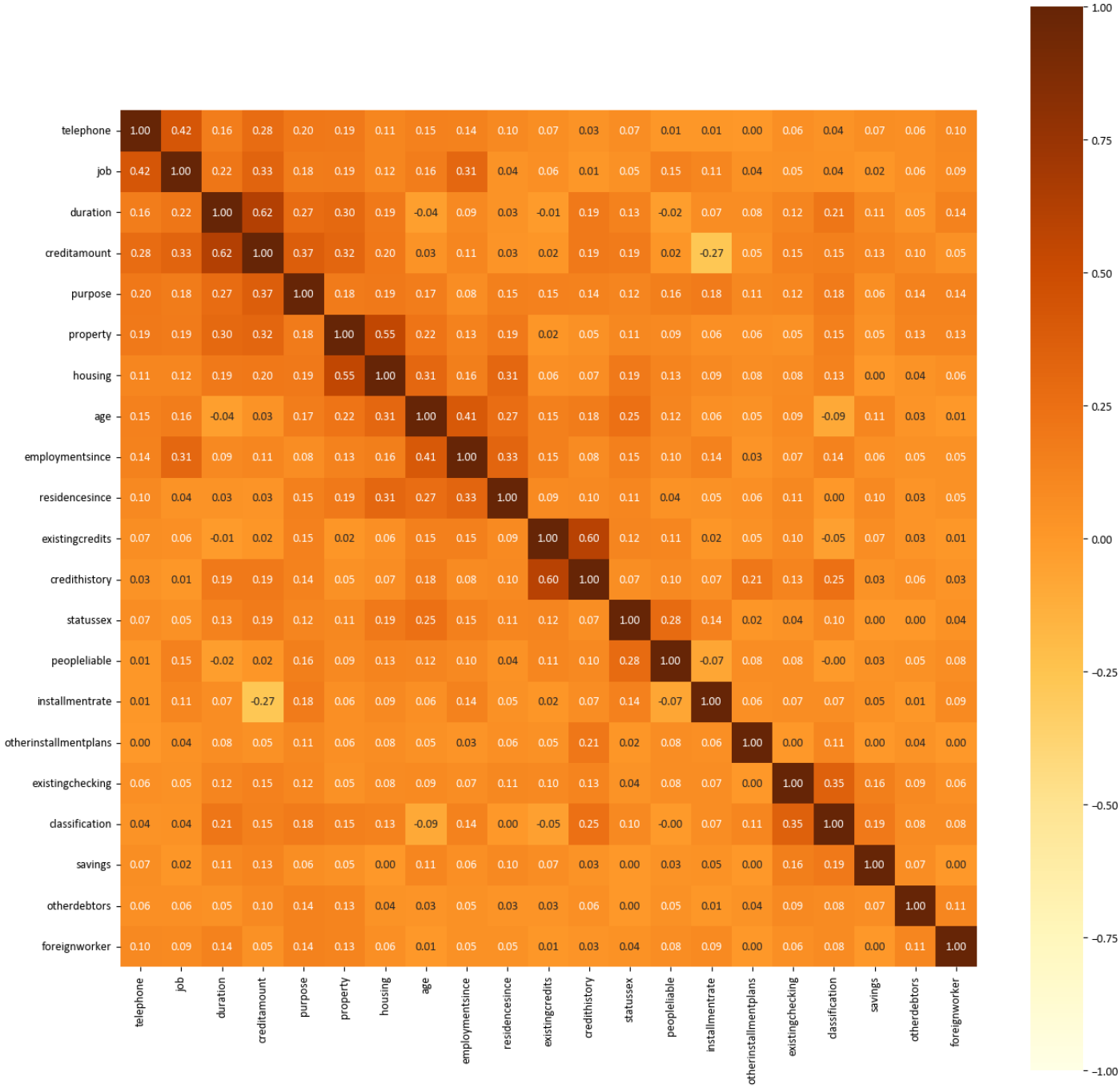
In [5]:
```
import fontTools.fontBuilder
import pandas as pd
import numpy as np
import matplotlib.pyplot as pt

## To remove warnings
import os
os.environ["OMP_NUM_THREADS"] = "4"
os.environ["LOKY_MAX_CPU_COUNT"] = "8"   # or your logical core count
import warnings
warnings.filterwarnings("ignore")
from networkx.classes import non_neighbors

names = ['existingchecking', 'duration', 'credithistory', 'purpose', 'creditamount',
         'savings', 'employmentsince', 'installmentrate', 'statussex', 'otherdebtors',
         'residencesince', 'property', 'age', 'otherinstallmentplans', 'housing',
         'existingcredits', 'job', 'peopleliable', 'telephone', 'foreignworker', 'classification'

customers = pd.read_csv('../../Potomac/AIT 620/LabWorks/PythonProject/datafiles/german.data', nar

customers.head()
```

Out[5]:

	existingchecking	duration	credithistory	purpose	creditamount	savings	employmentsince	installmer
0	A11	6	A34	A43	1169	A65	A75	
1	A12	48	A32	A43	5951	A61	A73	
2	A14	12	A34	A46	2096	A61	A74	
3	A11	42	A32	A42	7882	A61	A74	
4	A11	24	A33	A40	4870	A61	A73	

5 rows × 21 columns

In [17]:
```
from dython.nominal import associations
#fig = pt.figure(figsize=(16, 16))

pt.rcParams['font.family'] = 'Calibri'  # Example: use serif font family
pt.rcParams['font.size'] = 10         # Example: set default font size to 12

result = associations(customers, clustering=True, figsize=(16, 16),cmap='YlOrBr')
# Adjust tick label fonts

# Step 2 — get the axes AFTER associations() draws

pt.show()

catvars = ['existingchecking', 'credithistory', 'purpose', 'savings', 'employmentsince',
           'statussex', 'otherdebtors', 'property', 'otherinstallmentplans', 'housing', 'job',
           'telephone', 'foreignworker']
numvars = ['creditamount', 'duration', 'installmentrate', 'residencesince', 'age',
           'existingcredits', 'peopleliable', 'classification']

dummyvars = pd.get_dummies(customers[catvars])
transactions = pd.concat([customers[numvars], dummyvars], axis = 1)

transactions.head()
```



Out[17]:

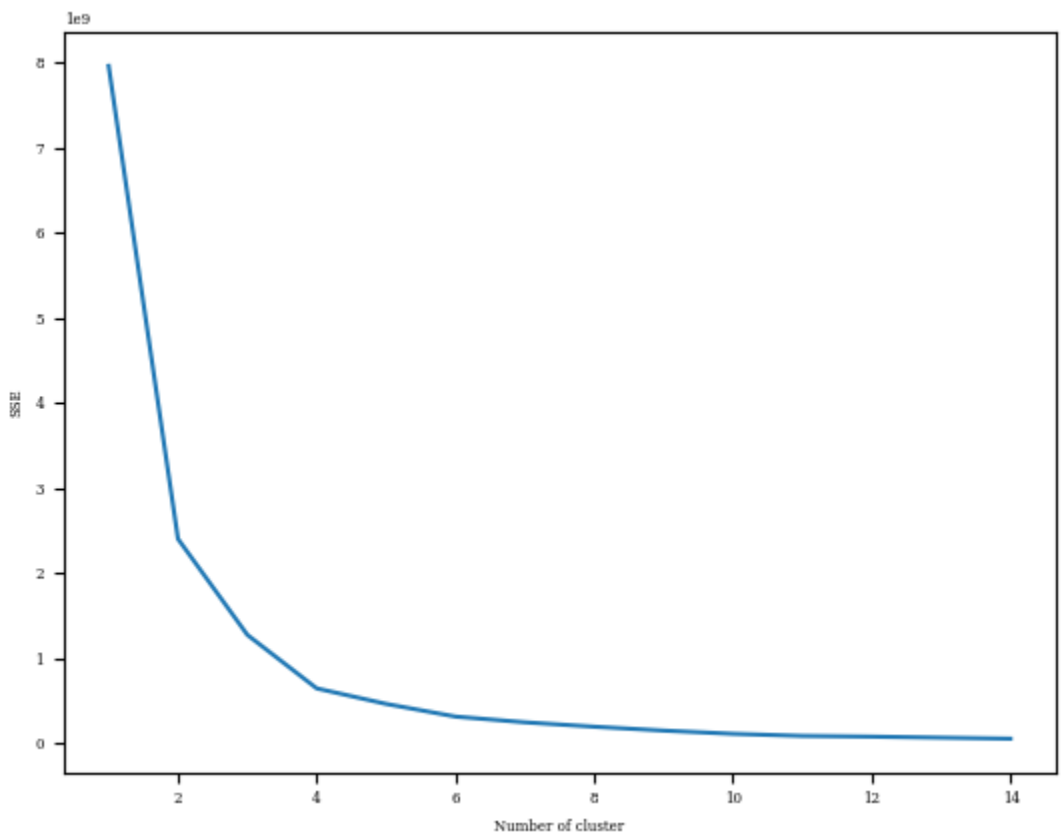
	creditamount	duration	installmentrate	residencesince	age	existingcredits	peopleliable	classificatio
0	1169	6	4	4	67	2	1	
1	5951	48	2	2	22	1	1	
2	2096	12	2	3	49	1	2	
3	7882	42	2	4	45	1	2	
4	4870	24	3	4	53	2	2	

5 rows × 62 columns

In [8]:

```
from sklearn.cluster import KMeans
from matplotlib import pyplot as plt
sse = {}
for k in range(1, 15):
    kmeans = (KMeans(n_clusters=k))
    kmeans.fit(transactions)
    sse[k] = kmeans.inertia_
```

```
plt.plot(list(sse.keys()), list(sse.values()))
plt.xlabel("Number of cluster")
plt.ylabel("SSE")
plt.show()
kmeans =(KMeans(n_clusters=4)).fit(transactions)
y = kmeans.labels_

clusters = (transactions.join(pd.DataFrame(data=y, columns=['cluster'])).
            groupby('cluster')
            .agg(
                age_mean=pd.NamedAgg(column='age', aggfunc='mean'),
                age_std=pd.NamedAgg(column='age', aggfunc='std'),
                creditamount=pd.NamedAgg(column='creditamount',aggfunc='mean'),
                duration=pd.NamedAgg(column='duration', aggfunc='mean'),
                count=pd.NamedAgg(column='age', aggfunc='count'),
                class_mean=pd.NamedAgg(column='classification', aggfunc='mean'),
                class_std=pd.NamedAgg(column='classification',aggfunc='std'),
            ).sort_values(by='class_mean'))
clusters
```



Out[8]:

cluster	age_mean	age_std	creditamount	duration	count	class_mean	class_std
0	35.631193	11.672843	1473.267890	15.099083	545	1.273394	0.446111
2	34.635088	10.738251	3603.540351	23.621053	285	1.277193	0.448400
1	36.798450	11.255124	7175.837209	33.232558	129	1.364341	0.483121
3	36.804878	11.914738	12576.463415	40.365854	41	1.609756	0.493865

In [9]:

```
from scipy.spatial.distance import pdist, squareform
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
```

```python
# Compute pairwise distances on scaled features
distances = squareform(
    pdist(
        StandardScaler().fit_transform(
            transactions[['classification', 'creditamount', 'duration']]
        )
    )
)

# Perform agglomerative clustering using the precomputed distance matrix
clustering = AgglomerativeClustering(
    n_clusters=5,
    metric='precomputed',
    linkage='average'
).fit(distances)

# Extract cluster labels
y = clustering.labels_

clusters = (
    transactions
        .join(pd.DataFrame(data=y, columns=['cluster']))
        .groupby(by='cluster')
        .agg(
            age_mean=pd.NamedAgg(column='age', aggfunc='mean'),
            age_std=pd.NamedAgg(column='age', aggfunc='std'),
            creditamount=pd.NamedAgg(column='creditamount', aggfunc='mean'),
            duration=pd.NamedAgg(column='duration', aggfunc='mean'),
            count=pd.NamedAgg(column='age', aggfunc='count'),
            class_mean=pd.NamedAgg(column='classification', aggfunc='mean'),
            class_std=pd.NamedAgg(column='classification', aggfunc='std'),
        )
        .sort_values(by='class_mean')
)

clusters
```

Out[9]:

cluster	age_mean	age_std	creditamount	duration	count	class_mean	class_std
4	35.920000	7.348025	8289.740000	42.640000	50	1.000000	0.000000
2	35.548165	11.489624	2473.405963	17.779817	872	1.259174	0.438433
0	38.200000	16.457352	15271.600000	51.300000	10	1.600000	0.516398
1	34.530303	10.989813	7845.363636	41.545455	66	2.000000	0.000000
3	45.500000	31.819805	14725.500000	6.000000	2	2.000000	0.000000

In [10]:

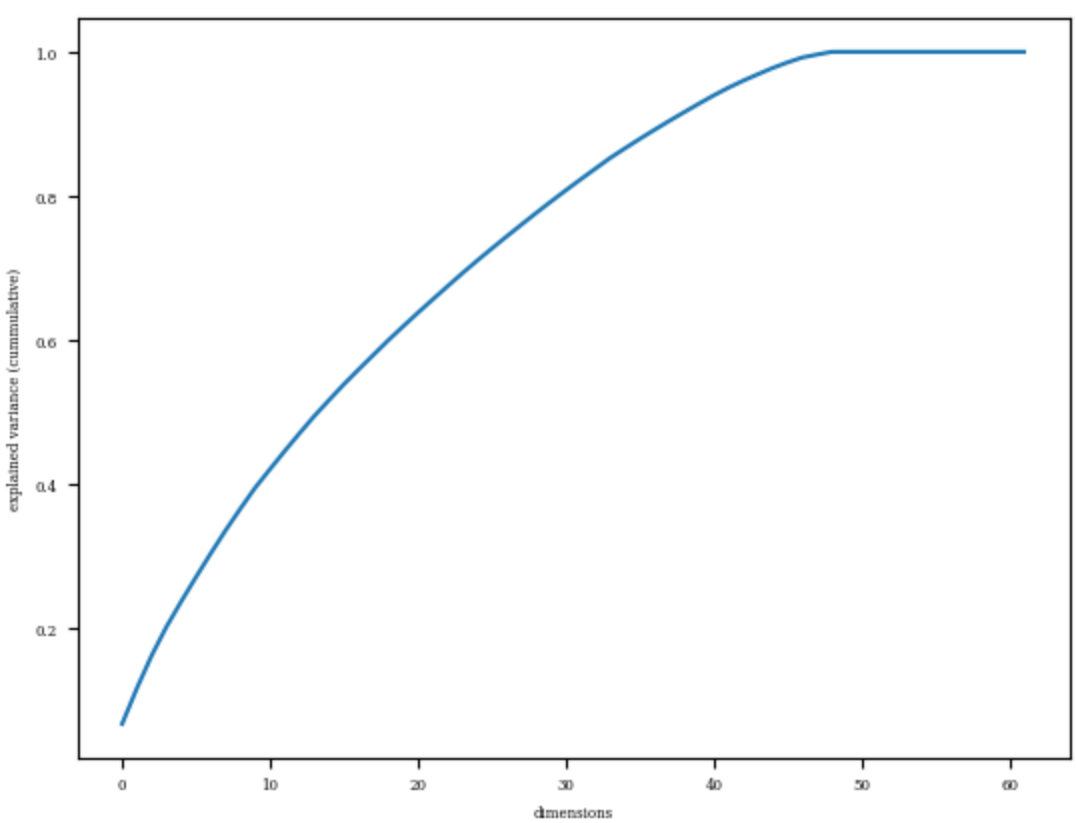
```python
import jax.numpy as jnp
import numpy as np

from jax import jit, vmap
from sklearn.base import ClassifierMixin

import jax
import random

from scipy.stats import hmean
```

```python
class KMeans(ClassifierMixin):

    def __init__(self, k, n_iter=100):
        self.k = k
        self.n_iter = n_iter

        # JIT-compiled batched Euclidean distance
        self.euclidean = jit(
            vmap(
                lambda x, y: jnp.linalg.norm(
                    x - y, ord=2, axis=-1, keepdims=False
                ),
                in_axes=(0, None),
                out_axes=0
            )
        )

    # def adjust_centers(self, X):
    #     return jnp.row_stack([
    #         X[self.clusters == c].mean(axis=0)
    #         for c in self.clusters
    #     ])

    def adjust_centers(self, X):
        return jnp.vstack([
            X[self.clusters == c].mean(axis=0)
            for c in range(self.k)
        ])

    def initialize_centers(self):
        """
        Roughly the k-means++ initialization
        """
        key = jax.random.PRNGKey(0)

        # JAX doesn't have uniform_multivariate
        self.centers = jax.random.multivariate_normal(
            key,
            jnp.mean(X, axis=0),
            jnp.cov(X, rowvar=False),
            shape=(1,)
        )

        for c in range(1, self.k):
            weights = self.euclidean(X, self.centers)

            if c > 1:
                weights = hmean(weights, axis=-1)
                print(weights.shape)

            # new_center = jnp.array(
            #     random.choices(X, weights=weights, k=1)[0],
            #     ndmin=2
            # )
            weights = self.euclidean(X, self.centers)  # shape (n_samples, n_centers)

            # Reduce to 1-D: distance to nearest center
            weights = weights.min(axis=1)

            # Convert distances to probabilities (invert so far points get higher weight)
            weights = 1 / (weights + 1e-8)
```

```
            # Normalize
            weights = weights / weights.sum()

            # Sample index
            idx = np.random.choice(len(X), p=np.array(weights))
            new_center = X[idx:idx + 1]

            self.centers = jnp.vstack((self.centers, new_center))
            print(self.centers.shape)

    def fit(self, X, y=None):
        self.initialize_centers()

        for iter in range(self.n_iter):
            dists = self.euclidean(X, self.centers)
            self.clusters = jnp.argmin(dists, axis=-1)
            self.centers = self.adjust_centers(X)


        return self.clusters
```

In [12]:

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA, FastICA
import matplotlib.pyplot as plt

pca = PCA().fit(
    StandardScaler().fit_transform(transactions)
)
ax = plt.plot(
    range(len(pca.explained_variance_ratio_)),
    np.cumsum(pca.explained_variance_ratio_)
)
plt.ylabel('explained variance (cummulative)')
plt.xlabel('dimensions')
plt.show()
```



In [13]:
```
len(transactions.columns)
```

Out[13]:
```
62
```

In [14]:
```
transactions.columns
```

Out[14]:
```
Index(['creditamount', 'duration', 'installmentrate', 'residencesince', 'age',
       'existingcredits', 'peopleliable', 'classification',
       'existingchecking_A11', 'existingchecking_A12', 'existingchecking_A13',
       'existingchecking_A14', 'credithistory_A30', 'credithistory_A31',
       'credithistory_A32', 'credithistory_A33', 'credithistory_A34',
       'purpose_A40', 'purpose_A41', 'purpose_A410', 'purpose_A42',
       'purpose_A43', 'purpose_A44', 'purpose_A45', 'purpose_A46',
       'purpose_A48', 'purpose_A49', 'savings_A61', 'savings_A62',
       'savings_A63', 'savings_A64', 'savings_A65', 'employmentsince_A71',
       'employmentsince_A72', 'employmentsince_A73', 'employmentsince_A74',
       'employmentsince_A75', 'statussex_A91', 'statussex_A92',
       'statussex_A93', 'statussex_A94', 'otherdebtors_A101',
       'otherdebtors_A102', 'otherdebtors_A103', 'property_A121',
       'property_A122', 'property_A123', 'property_A124',
       'otherinstallmentplans_A141', 'otherinstallmentplans_A142',
       'otherinstallmentplans_A143', 'housing_A151', 'housing_A152',
       'housing_A153', 'job_A171', 'job_A172', 'job_A173', 'job_A174',
       'telephone_A191', 'telephone_A192', 'foreignworker_A201',
       'foreignworker_A202'],
      dtype='str')
```

In [15]:
```
from sklearn.manifold import LocallyLinearEmbedding
import numpy as np

dimensions_range = [1, 11]   # n_components = 1..10
n_neighbors = 10             # must be >= max n_components

reconstruction_errors = np.zeros(dimensions_range[1])

for n_components in range(*dimensions_range):

    lle = LocallyLinearEmbedding(
        n_components=n_components,
        n_neighbors=n_neighbors,
        n_jobs=-1,
        method='standard'
    ).fit(transactions)

    # LLE always provides the barycentric weight matrix

    error = lle.reconstruction_error_
    reconstruction_errors[n_components] = error
print("Reconstruction errors:", reconstruction_errors[1:11])
```

```
Reconstruction errors: [9.60310428e-11 6.99560259e-09 2.13892850e-08 7.59813195e-08
 2.09924977e-07 4.51334297e-07 1.04707349e-06 1.91063371e-06
 3.45998696e-06 5.69324003e-06]
```

In [ ]: